ONE SOURCE VIDEO · VGGT DEPTH · FREE ROAMING

A classroom from one continuous video

We reconstructed a classroom from **C5306.MP4 only**. The browser renders a 3D Gaussian Splat. Visitors can walk through the captured aisle, look around, and follow a short route to the front of the room.

198 / 198
sampled frames in one camera model

873,736
Gaussian splats in a 26.66 MiB browser asset

20,000
Splatfacto training steps on the NST GPU node

Unknown
physical room dimensions; scale is relative

System architecture

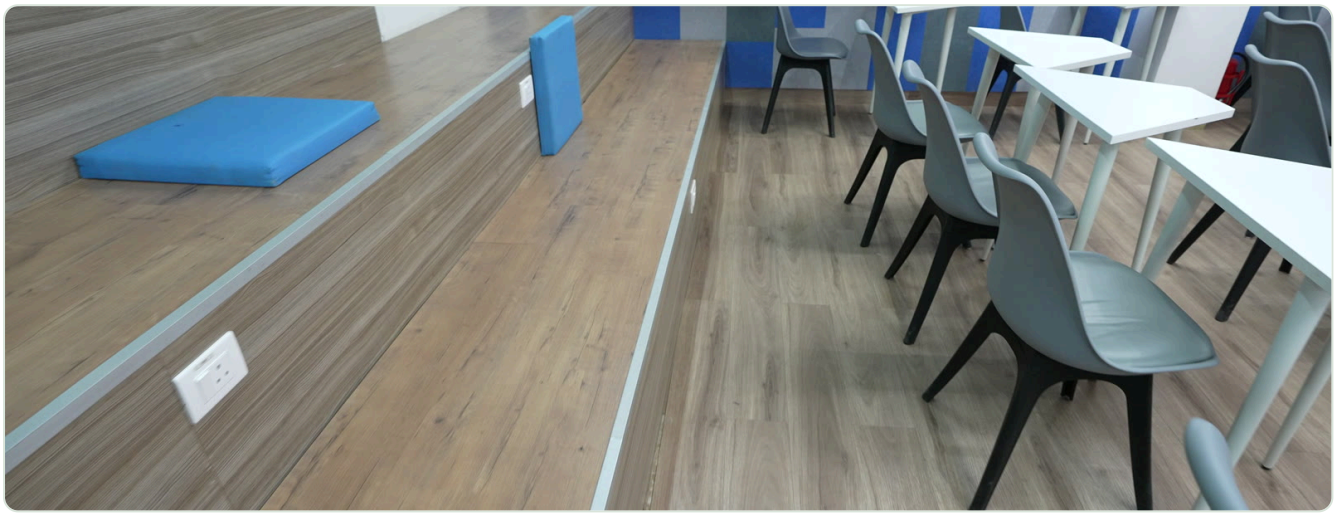


The app uses Next.js, React, Three.js, and a Gaussian splat renderer. Scene metadata sets the start view, relative scale, and preview walk limits. Rendering and navigation logic are separate.

> **Scope.** This is a visual reconstruction with a walking preview. The splat is not a measured room mesh. No furniture collision is claimed for this model. The video has limited views of the upper walls and ceiling.

Record a stable, overlapping room pass

The repeat-capture SOP has seven actions. Select one room. Record one continuous video with one lens. Keep people out and use steady light. Walk slowly with translation. Show walls, floor edges, ceiling, and furniture from several positions. Avoid fast pans. Keep the raw video and checksum. Check that the camera route is continuous before training.



Frame 80 from C5306.MP4 shows the stage steps, floor, desks, and chairs. The camera faces down, which limits upper-room coverage.

Actual source and frame selection

Input	`assets/C5306.MP4`, Sony H.264 video, 3840 × 2160, 50 fps, 49.44 s
Integrity	SHA-256 `9e4124bb4c7d8dd19fb31de78796f71b932256dfaacd230aeb2a75cb00c54972`
Selection	Sample the full video at 4 fps. This gives 198 frames. Use sequential feature matching at the source resolution.
Camera route	All 198 frames registered in one model. The maximum adjacent camera step was 1.22 times the median step.

The video was supplied before this reconstruction. We did not control its original capture. We used no still photos, other video, prior camera poses, or measured room size. The source points down for much of the route. The model cannot show a complete ceiling from this recording.

Validate depth against the video camera solve

Sequential PyCOLMAP reconstruction registered 198 of 198 frames in one model with 59,929 sparse points. Its mean reprojection error was 0.754 pixels. The median adjacent camera step was 0.141 model units and the maximum was 0.171.

VGGT inferred depth and camera predictions for 25 sampled frames. We fitted each depth map to sparse points from the same video. All 25 had a positive depth scale. Median relative depth error was 1.14% at matched sparse points. VGGT camera poses were less consistent, so training used PyCOLMAP poses and VGGT depth points.

Training and export

1. Filter the sparse cloud to 58,344 starting points. Add 51,800 depth points that pass confidence and room-envelope checks.
2. Train Nerfstudio Splatfacto for 20,000 steps on the NST GPU node. Disable camera optimization and scale regularization.
3. Fit the visible floor using the same video. Define the median camera clearance as one arbitrary scene unit.
4. Remove distant outliers and export 873,736 records in the standard 32-byte `.splat` layout.

Relative scale
The fitted floor sets Y = 0. No physical length was measured. Positions and routes use scene units.

Browser result
The 26.66 MiB asset loads in the desktop viewer. The captured aisle supports free roaming and a short front route.

Visual limits

The floor, stage steps, desks, and chairs are recognizable. A dark patch remains near the upper front wall. The ceiling and upper walls have weak coverage because the camera points down. Extra training cannot restore surfaces that the source did not record.

> Held-out image scores are PSNR 31.05, SSIM 0.929, and LPIPS 0.174. Test images are nearby frames from the same video, so these scores can overstate quality from new viewpoints. There is no measured room size or mobile performance result.

Walk through the captured aisle

The C5306 room is the app's default scene. Visitors use WASD or arrow keys to move, and the mouse to look. The viewer streams the splat, shows load progress, and has a reset control. It reports camera positions and route lengths in scene units.

Search finds one front-aisle point of interest. Its route stays near recorded camera positions. Walk limits are visual estimates. The app does not claim verified furniture collision for this scene.

Reproduce and inspect

The repository contains the frame and camera reconstruction pipeline, VGGT depth inference and validation, floor calibration, and splat export scripts. The C5304 model remains available as an earlier comparison scene.

```
npm ci → npm test → npm run build → npm run start
```

Open the site in a WebGL browser. The full reconstruction needs the original C5306.MP4 source, which is not stored in Git.

Open-source tools and credits

Camera reconstruction: COLMAP / PyCOLMAP. Depth: VGGT. Gaussian training: Nerfstudio Splatfacto and gsplat. Viewer: Next.js, React, Three.js, and GaussianSplats3D. Media extraction: FFmpeg. Each tool retains its own license.

Access and review

The model, viewer, and matching code branch are available locally. This copy has no public viewer link. The demonstration video shows the C5306 source, the reconstruction method, the live viewer, and its current coverage limits.